

SIMATS ENGINEERING

Department of Computer Science & Engineering

ASSESSMENT TOOL 1- CONSTRAINT-BASED PROBLEM SOLVING

Course Code:	CSA12	Course Title:	Computer Architecture
CO Assessed:	CO5	Assessment:	ASSESSMENT TOOL1- Constraint-Based Problem Solving
Weightage:	40%	Total Marks:	25
CO5 Statement:	Construct I/O communication mechanisms by comparing memory-mapped and I/O-mapped techniques, interrupt handling (vectored, hardware, software), DMA controllers, and advanced bus interfaces including PCIe, USB, NVMe, and Thunderbolt.		
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Department:	B.Tech AITM	Date:	26/08/26

ANNEXURE A

Constraint-Based Problem Solving

1. Construct an I/O communication mechanism for a real-time embedded system with the following constraints:

- CPU utilization must be below 15%
- Multiple sensors generate interrupts every 1 ms
- Use vectored interrupt handling
- Select between memory-mapped and I/O-mapped I/O for optimal latency

2. Design an I/O subsystem under these constraints:

- High-speed storage requires throughput > 3 GB/s
- Must use NVMe over PCIe
- CPU should not handle bulk data transfer
- Integrate DMA controller with interrupt notification

3. Construct a multi-device communication architecture with constraints:

- 6 peripherals using USB, 2 high-speed devices using Thunderbolt
- Avoid interrupt starvation
- Use hardware and software interrupts
- Ensure fair bus arbitration

4. Design a data acquisition system with constraints:

- Continuous data stream at 500 MB/s
- Minimal CPU intervention required
- Must compare DMA transfer vs interrupt-driven I/O
- Use memory-mapped I/O for device registers

5. Construct an interrupt handling mechanism with constraints:

- Support nested interrupts
- Maximum interrupt latency < 10 μ s
- Use vectored interrupts for high-priority devices
- Use software interrupts for background tasks

78%

Code of Conduct:

I V. Adarsh Reddy (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

V. Adarsh Reddy
Signature of the student:

MARKS ALLOCATION

	Problem Understanding (25%)	Constraint Analysis (25%)	Solution Development (25%)	Application of Concepts (15%)	Justification (10%)
Q1	2	2	2	1	1
Q2	2	1.5	2	1	0.5
Q3	2.5	1.5	2.5	1.5	0.5
Q4	2	2	2	1	0.5
Q5	2	2	2	1	1

RUBRICS FOR CALCULATION:

39

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Problem Understanding	25% 2.5	Clearly defines the problem and identifies all relevant constraints accurately	Identifies most constraints with minor gaps	Partial understanding; misses key constraints	Fails to identify constraints correctly
Constraint Analysis	25% 2.5	Analyzes constraints effectively and prioritizes them logically	Provides reasonable analysis with some prioritization	Limited analysis; weak prioritization	No meaningful analysis or prioritization
Solution Development	25% 2.5	Develops optimal and feasible solutions satisfying all constraints	Develops workable solutions with minor limitations	Solutions partially satisfy constraints	Solutions do not meet constraints
Application of Concepts	15% 1.5	Effectively applies appropriate concepts, methods, or tools	Applies concepts with minor errors	Limited application; requires guidance	Unable to apply relevant concepts